JONAH MERRIAM

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EDUCATION

University of Connecticut August 2024 – May 2027
BSE in Computer Science and Engineering (Systems and Networks) — GPA: 3.54 / 4.00 *Graduating One Year Early*

RESEARCH

BGPy with C Extensions — Undergraduate Research, University of Connecticut Jan 2025 – Present

- Co-authoring BGPyC: A BGP Python Security Simulator with C Extensions, submitted to CSET 2026; supervised by Prof. Furuness.
- Accelerated critical simulation paths by integrating C-style Python objects; improved memory efficiency and routing experiment throughput. github.com/jonahmer22/bgpy_pkg

PROJECTS

Forth Interpreter Jun 2026 – Present

- Built a Forth 2012 core-compliant interpreter in C featuring indirect threaded code (ITC) execution, a linked-list dictionary, and a custom assembler for primitive words. github.com/jonahmer22/forth
- Implemented the full standard word set including control flow, string handling, and defining words; supports both interactive REPL and file-based execution modes.

Cortex-VM Jan 2026 – Present

- Designed and implemented a general-purpose 64-bit RISC-V-inspired virtual machine in C, achieving ~400M operations/second on a single core. github.com/jonahmer22/cortex-vm
- Built a custom 64-bit ISA with a three-arena memory model, defined calling convention, and binary file format; serves as the primary compilation target for Marrow-Lang.

ReMem Aug 2025 – Present

- Built a high-performance memory allocator in C featuring stack-walking garbage collection, size-classed arenas, and O(1) page indexing. github.com/jonahmer22/ReMem
- Designed as a standalone reusable submodule with a clean API; intended for integration into C projects requiring efficient memory reuse.

Marrow-Lang Sep 2025 – Present

- Designing a compiled/interpreted language targeting Cortex-VM with emphasis on performance and low-level control. github.com/jonahmer22/marrow-lang
- Implemented tokenizer and expression evaluator in C; planning full compiler pipeline with custom code generation targeting Cortex ISA.

BGP Simulator Aug 2025 – Present

- Implemented a Border Gateway Protocol simulator in C emphasizing runtime efficiency and correct protocol behavior. github.com/jonahmer22/bgp-simulator
- Achieved one of the fastest runtimes in the course through careful memory management and algorithmic optimization.

PUBLICATIONS

BGPyC: A BGP Python Security Simulator with C Extensions *Submitting to CSET 2026*

WORK EXPERIENCE

Service Clerk, Big Y Market Aug 2023 – Present

- Maintained consistent performance in a customer-facing role while managing full-time academic coursework and undergraduate research commitments.

AWARDS

Rochester Institute of Technology Computing Award May 2023

SKILLS

Programming: C, C++, Python, Assembly (RISC-V), Java, Forth, SQL, HTML/CSS/JS, GTest, PyTest, Python packaging
Infrastructure: Linux (Debian), Bash, Git & GitHub Actions, self-hosted server administration, Cloudflare, Pi-hole
Tools: Vim, Tmux, LaTeX, Jira, Markdown